Salt and hypertension[1]

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I am pleased to address your group, because it is heavily oriented toward diseases of infancy and childhood. It has long been my belief that the most prevalent cardiovascular diseases, atherosclerosis and hypertension, are generated early in life rather than some 4 or 5 decades later when they are commonly diagnosed. I will consider myself successful if I engage your permanent interest in this concept relative to hypertension, which I have been studying exclusively for more than 2 decades. I will remain unperturbed, however, if you do not accept all the details.

During the 1st decade, our work was focused on the role of sodium in the etiology and pathogenesis of clinical and experimental hypertension. In the 2nd decade, the importance of the genetic substratum was recognized and studied experimentally. In this 3rd decade, we are trying to define the genetic factors responsible for hypertension and to quantify individual genetic vulnerability to this disease. In this review of our work, I will let you judge its overall relevance.

Our work began in 1948 and was based on Kempner's indubitable results with the rice–fruit diet in the treatment of hypertension (1, 2). For reasons that are difficult to fathom, there appeared a great deal of antipathy to Kempner's reports as well as irrational disbelief in the effectiveness of the diet. In a major review of the subject early in 1949, it was stated that "suggestion" had not been ruled out and that "Restriction of salt intake has not been conclusively shown to be independently capable of lowering the blood pressure in hypertensive patients" (3).

We judged Kempner's work only on the basis of his quantitative evidence. We were also aware of several physicians, incapacitated by progressive hypertensive cardiovascular disease who, after being on the regimen for some months, were back practicing medicine. We found it believable that the rice–fruit diet was often useful in the treatment of hypertension, including the "malignant" phase. In so concluding, I have often felt that we became heirs to the antipathies originally directed at Allen and later at Kempner. We decided, nonetheless, to try to detect the dietary factor that made this diet effective. For those of you who are unfamiliar with the diet, let me define it as a low sodium, or a high potassium, or a high carbohydrate, or a low protein, or a low fat, or a high fluid diet. In its pristine form it is made up mostly of rice and fruit including juices, but certainly with no added salt.

In a series of careful metabolic studies carried out over 4 years at the Rockefeller Institute with Drs. Dole and Cotzias, we showed that the blood pressure lowering effect was due primarily to its low sodium content (4–6). Parallel studies at the nearby Goldwater Memorial Hospital in New York City by Watkin et al. (7) and Hatch et al. (8) resulted in similar conclusions.

As a direct outcome of these studies, I began to wonder about the following: if a low salt intake causes a lowering of blood pressure, does a high salt intake cause increased blood pressure? Both the groups mentioned had found that the addition of NaCl to the rice–fruit diet prevented its therapeutic effect, whereas additions of NaCl after the induction of therapy caused reemergence of hypertension. This was not quite the answer to the question that I was asking and which I thought to be original at the time. Later, however, it was learned that several others had had similar thoughts much earlier, namely Ambard and Beaujard in

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